Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The fundamental frequency of an organ pipe is $512\ \text{Hz}$. If its length is increased, then frequency will

(A) decrease

(B) increase

(C) remains same

(D) cannot be predicted

Q.2

A closed organ pipe (closed at one end) is excited to support the third overtone. It is found that air in the pipe has

(A) three nodes and three antinodes

(B) three nodes and four antinodes

(C) four nodes and three antinodes

(D) four nodes and four antinodes

Q.3

A small piece of cork in a ripple tank oscillates up and down as ripple pass it. If the ripples travelling at $0.3\ \text{m/s}$ have a wavelength of $1.5\pi\text{cm}$ and the cork vibrates with an amplitude of $5\ \text{mm}$, the maximum velocity of the cork is:

(A) $20\ \text{cm/sec}$

(B) $20\ \text{m/sec}$

(C) $0.02\ \text{m/sec}$

(D) $200\ \text{cm/sec}$

Q.4

The fundamental frequency of a closed end organ pipe is n. Its length is doubled and radius is halved. Its frequency will become nearly

(A) $\text{n}/2$

(B) $\text{n}/3$

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(C) n

(D) $2 \mathrm{n}$

Q.5

Tube A has both ends open while tube B has one end closed, otherwise they are identical. The ratio of fundamental frequency of tube A and B is

(A) $1:2$

(B) $1:4$

(C) $2:1$

(D) $4:1$

Q.6

The number of possible natural oscillation of air column in a pipe closed at one end of length $85 \mathrm{~cm}$ whose frequencies lie below $1250 \mathrm{~Hz}$ are (velocity of sound $=340 \mathrm{~ms}^{-1}$)

(A) 4

(B) 5

(C) 7

(D) 6

Q.7

The fundamental frequency of a closed organ pipe of length $20 \mathrm{~cm}$ is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is

(A) $100 \mathrm{~cm}$

(B) $120 \mathrm{~cm}$

(C) $140 \mathrm{~cm}$

(D) $80 \mathrm{~cm}$

Q.8

An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at both ends vibrating in third overtone are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is

(A) $8/3$

(B) $3/8$

(C) $1/2$

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(D) $1/3$

Q.9

The fundamental frequency of an open organ pipe is $300\ \mathrm{Hz}$. The first overtone of this pipe has same frequency as first overtone of a closed organ pipe. If speed of sound is $330\ \mathrm{m/s}$, then the length of closed organ pipe is

(A) $41\ \mathrm{cm}$

(B) $37\ \mathrm{cm}$

(C) $31\ \mathrm{cm}$

(D) $80\ \mathrm{cm}$

Q.10

While measuring the speed of sound by performing a resonance column expenment, a student gets the first resonance condition at a column length of $18\ \mathrm{cm}$ during winter. Repeating the same experiment during summer, she measures the column length to be $x\ \mathrm{cm}$ for the second resonance. Then

(A) $x > 54$

(B) $54 > x > 36$

(C) $36 > x > 18$

(D) $18 > x$

Q.11

When two sound waves travel in the same direction in a medium, the displacements of a particle located at ' x ' at time ' t ' is given by :

$$y_1 = 0.05\cos(0.50\pi x - 100\pi t)$$
$$y_2 = 0.05\cos(0.46\pi x - 92\pi t)$$

where y_1, y_2 and x are in meters and t in seconds. The speed of sound in the medium is :

(A) $92\ \mathrm{m/s}$

(B) $200\ \mathrm{m/s}$

(C) $100\ \mathrm{m/s}$

(D) $332\ \mathrm{m/s}$

Q.12

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The extension in a string, obeying Hooke's law, is x. The speed of sound in the stretched string is v. If the extension in the string is increased to $1.5x$, the speed of sound will be

(A) $1.22\mathrm{v}$

(B) $0.61\mathrm{v}$

(C) $1.50\mathrm{v}$

(D) $0.75\mathrm{v}$

Q.13

Sound waves are travelling in a medium whose adiabatic elasticity is E and isothermal elasticity E'. The velocity of sound waves is proportional to

(A) E'

(B) $\sqrt{\mathrm{E}}$

(C) $\sqrt{\mathrm{E}'}$

(D) $\frac{E}{E'}$

Q.14

What is the effect of humidity on sound waves when humidity increases?

(A) Speed of sound waves is more

(B) Speed of sound waves is less

(C) Speed of sound waves remains same

(D) Speed of sound waves becomes zero

Q.15

The bulk modulus of a liquid of density $8000\ \mathrm{kg\ m^{-3}}$ is $2 \times 10^{9}\ \mathrm{N\ m^{-2}}$. The speed of sound in that liquid is (in $\mathrm{ms^{-1}}$)

(A) 200

(B) 250

(C) 100

(D) 500

Q.16

The velocity of sound in hydrogen is $1224\ \mathrm{m/s}$. Its velocity in a mixture of hydrogen and oxygen containing 4 parts by volume of hydrogen and 1
part oxygen is

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(A) $1224\ \mathrm{m/s}$

(B) $612\ \mathrm{m/s}$

(C) $2448\ \mathrm{m/s}$

(D) $306\ \mathrm{m/s}$

Q.17

A pipe of length ℓ_1, closed at one end is kept in a chamber of gas of density ρ_1. A second pipe open at both ends is placed in a second chamber of gas of density ρ_2. The compressibility of both the gases is equal. Calculate the length of the second pipe if frequency of first overtone in both the cases is equal

(A) $\frac{4}{3}\ell_1\sqrt{\frac{\rho_2}{\rho_1}}$

(B) $\frac{4}{3}\ell_1\sqrt{\frac{\rho_1}{\rho_2}}$

(C) $\ell_1\sqrt{\frac{\rho_2}{\rho_1}}$

(D) $\ell_1\sqrt{\frac{\rho_1}{\rho_2}}$

Q.18

When a sound wave goes from one medium to another, the quantity that remains unchanged is

(A) frequency

(B) amplitude

(C) wavelength

(D) speed

Q.19

A progressive sound wave of frequency $500\ \mathrm{Hz}$ is travelling through air with a speed of $350\ \mathrm{ms}^{-1}$. A compression maximum appears at a place at a given instant. The minimum time interval after which the rarefraction maximum occurs at the same point, is

(A) $200\ \mathrm{s}$

(B) $\frac{1}{250}\ \mathrm{s}$

(C) $\frac{1}{500}\ \mathrm{s}$

(D) $\frac{1}{1000}\ \mathrm{s}$

Q.20

A person speaking normally produces a sound intensity of $40\ \mathrm{dB}$ at a distance of $1\ \mathrm{m}$. If the threshold intensity for reasonable audibility is $20\ \mathrm{dB}$, the maximum distance at which he can be heard clearly is

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(A) $4\ \mathrm{m}$

(B) $5\ \mathrm{m}$

(C) $10\ \mathrm{m}$

(D) $20\ \mathrm{m}$

Answers & Solutions

Q.1 Answer:

decrease

Solution:

Frequency of Closed Organ pipe
$\eta = \frac{V}{4L}$
$L \longrightarrow$ **length of losed organ pipe**
$\eta \rightarrow$ **fundamental frequency**
(frequency) $\eta \propto \frac{1}{L}$.
hence as length increased frequency (η) **decreases**

Q.2 Answer:

four nodes and four antinodes

Solution:

Third overtone has a frequency $7\mathrm{n}$**, which means** $L = \frac{7\lambda}{4} =$ **three full loops** $+$ **one half loop, which would make four nodes and four antinodes.**

Q.3 Answer:

$20\ \mathrm{cm/sec}$

Solution:

Ginen:
velocity of ripples, $v_r = 0:3\ \mathrm{m/s} \Rightarrow v_r = 30\ \mathrm{cm/s};$
Wavelength of ripples, $\lambda_r = 1.5\pi \mathrm{cm};$
Amplitude of cork, $a = 5\mathrm{mn} \Rightarrow a = 0.5\ \mathrm{cm}$
Frequency of ripples, $D_r = \frac{v_r}{\lambda r} = \frac{30}{1.5\pi} \Rightarrow \nu_r = \frac{20}{\pi} H_z$
maximum velocity of cork, $V_{\max} = 0.\omega$ $\omega \rightarrow$ **angular frequency** $= 2\pi D$

$$\text{So,} \quad V_{\max} = 0.5\left(2\pi\left(\frac{20}{\pi}\right)\right)$$
$$\Rightarrow V_{\max} = 20\ \mathrm{cm/s}$$

Q.4 Answer:

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$n/2$

Solution:

Fundamental frequency of the closed end organ pipe is

$$v_{0c} = \frac{v}{4(l + 0.6r)}$$

After its length is doubled and radius is halved, the fundamental frequency becomes

$$v'_{0c} = \frac{v}{4(2l + 0.6r/2)}$$

Thus, $\frac{v_{0c}}{v'_{0c}} = \frac{4(2l+0.6r/2)}{4(l+0.3r)}$

Since, $1 >> r$

$$\frac{v_{0c}}{v'_{0c}} = 2$$

$$v'_{0c} = n/2$$

Q.5 Answer:

$2 : 1$

Solution:

The frequency of sound in open organ pipe is:

$$n_A = \frac{v}{2l}$$

The frequency of sound in closed organ pipe is: $n_B = \frac{v}{4l}$

Therefore, the ratio of the frequencies is: $\Rightarrow n_A/n_B = 2 : 1$

Q.6 Answer:

6

Solution:

For pipe closed at one end,

$$f_n = n\left(\frac{v}{4l}\right)$$

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where n is the odd number

$$f_n = n\left[\frac{340}{4 \times 85 \times 10^{-2}}\right]$$
$$= n[100]$$

N is an odd number, so for the given condition, the value of n are:

$$n = 1, 3, 5, 7, 9, 11$$

So, the number of possible natural oscillations could be 6 .

Q.7 Answer:

120 cm

Solution:

For closed organ pipe, fundamental frequency is given by $v_c = v/4l$ For open organ pipe, fundamental frequency is given by, $\mathrm{v_0} = \mathrm{v}/2\mathrm{l}$ ' 2nd overtone of open organ pipe

$$v' = 3v_o; v' = \frac{3v}{2l'}$$

According to question, $v_c = v'$

$$\frac{v}{4l} = \frac{3v}{2l'}$$
$$l' = 6l$$

Here, $l = 20 \text{ cm}, l' = ?$

$\therefore \quad l' = 6 \times 20 = 120 \text{ cm}$

Q.8 Answer:

$3/8$

Solution:

First overtone of closed organ pipe:

$$\frac{3\lambda}{4} = \mathrm{L_1}$$

$$\mathrm{f_1} = \frac{\mathrm{v}}{\lambda_1} = \frac{3\mathrm{v}}{4\ \mathrm{L_1}}$$

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Third overtone of open organ pipe:

$$\frac{4\lambda}{2} = \mathrm{L}_2$$

$$\mathrm{f}_2 = \frac{\mathrm{v}}{\lambda_2} = \frac{2\mathrm{v}}{\mathrm{L}_2}$$

Given $\mathrm{f}_1 = \mathrm{f}_2$

$$\frac{3\mathrm{v}}{4\,\mathrm{L}_1} = \frac{2\mathrm{v}}{\mathrm{L}_2}$$

$$\Rightarrow \frac{\mathrm{L}_1}{\mathrm{L}_2} = \frac{3}{8}$$

Q.9 Answer:

41 cm

Solution:

If l_1 and l_2 are lengths of open and closed organ pipes then

(i) first overtone of organ pipe $2v/2l_1$,

(ii) first overtone of closed pipe $3v/4l_2$

(iii) Fundamental frequency of open organ pipe $= \mathrm{v}/2l_1 = 300$

$$\Rightarrow \frac{330}{2l_1} = 300 \Rightarrow l_1 = \left(\frac{330}{300 \times 2}\right) m = 55 \text{ cm}$$

By second condition $\frac{2v}{2l_1} = \frac{3v}{4l_2}$

$$\Rightarrow \tfrac{1}{l_1} = \tfrac{3}{4l_2} \Rightarrow l_2 = \tfrac{3}{4} \times l_1$$
$$= \tfrac{3}{4} \times 55 \text{ cm} = 41.3 \text{ cm}$$

Q.10 Answer:

$x > 54$

Solution:

$\frac{\lambda_w}{4} = 18$ cm

$\lambda_w = 72$ cm

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For second resonance $\frac{3\lambda_s}{4} = x$

as velocity increased in summer $\Rightarrow \lambda_s > \lambda_w$

$\Rightarrow x > \frac{3\times72}{4}$

$x > 54$ cm

Q.11 Answer:

200 m/s

Solution:

Standard equation $y(x,t) = A\cos\left(\frac{\omega}{V}x - \omega t\right)$

From any of the displacement equation Say y_1

$\frac{\omega}{V} = 0.50\pi$ **and** $\omega = 100\pi$

$\therefore \frac{100\pi}{V} = 0.5\pi$

$\therefore \quad V = \frac{100\pi}{0.5\pi} = 200$ m/s

Q.12 Answer:

$1.22v$

Solution:

According to Hooke's law $F_R \propto x$ **[Restoring force** $F_R = T$**, tension of spring] Velocity of sound by a stretched string** $v = \sqrt{\frac{T}{m}}$ **where** m **is the mass per unit length Hence** $v \propto \sqrt{T}$ **or,** $\frac{v}{v'} = \sqrt{\frac{T}{T'}}$ **or** $v' = v\sqrt{\frac{T'}{T}} = v\sqrt{\frac{1.5x}{x}} = 1.22v$

Q.13 Answer:

$\sqrt{E}$

Solution:

$V = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{E}{\rho}}$

Q.14 Answer:

Speed of sound waves is more

Solution:

Velocity of sound $= \sqrt{\frac{\gamma RT}{M}}$ **When water vapour are present in air, average molecular weight of air decreases and hence velocity increases.**

Q.15 Answer:

500

Solution:

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The speed of sound in liquid, $\mathrm{v}=\sqrt{\frac{\mathrm{k}}{\rho}}=\sqrt{\frac{2\times10^{9}}{8000}}=\sqrt{\frac{1}{4}\times10^{6}}$

$\mathrm{v}=\frac{1}{2}\times10^{3}=500\ \mathrm{m\ s^{-1}}$.

Q.16 Answer:

$612\ \mathrm{m/s}$

Solution:

If $\rho_{\mathrm{H}}=1$**, then** $\rho_{\mathrm{mix}}=\frac{4\times1+1\times16}{(4+1)}=4$

$\frac{\mathrm{v_{mix}}}{\mathrm{v_H}}=\sqrt{\frac{\rho_{\mathrm{H}}}{\rho_{\mathrm{mix}}}}=\sqrt{\frac{1}{4}}=\frac{1}{2}$

$\mathrm{v_{mix}}=\frac{\mathrm{v_H}}{2}=\frac{1224}{2}=612\ \mathrm{m/s}$

Q.17 Answer:

$\frac{4}{3}\ell_1\sqrt{\frac{\rho_1}{\rho_2}}$

Solution:

Frequency of $\mathrm{I^{st}}$ **of over tone of closed pipe = Frequency of** $\mathrm{I^{st}}$ **of over tone of open pipe**

$$\Rightarrow\frac{3\mathrm{v}}{4l_1}=\frac{\mathrm{v}}{\mathrm{l_2}}$$

$$\Rightarrow\frac{3}{4l_1}\sqrt{\frac{\gamma\mathrm{P}}{\mathrm{P}_1}}=\frac{1}{\mathrm{l_2}}\sqrt{\frac{\gamma\mathrm{P}}{\mathrm{P}_2}}$$

$$\therefore\mathrm{l_2}=\frac{4}{3}\mathrm{l}_1\sqrt{\frac{\mathrm{P}_1}{\mathrm{P}_2}}$$

Q.18 Answer:

frequency

Solution:

Frequency of wave is a function of the source of waves. Therefore, it remains unchanged.

Q.19 Answer:

$\frac{1}{1000}\ \mathrm{s}$

Solution:

$\mathrm{T}=\frac{1}{500}\ \mathrm{s};\mathrm{t}=\frac{\mathrm{T}}{2}=\frac{1}{1000}\ \mathrm{s}$

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Q.20 Answer:

$10\ \mathrm{m}$

Solution:

Loudness in decibels is given by: $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$

Where I_0 **= threshold intensity of sound** $= 10^{-12}\ \mathrm{W/m^2}$

i.e., $40 = 10\log_{10}\left(\frac{I_1}{I_0}\right)$ ..... **(i)**

and $20 = 10\log_{10}\left(\frac{I_2}{I_0}\right)$ ..... **(ii)**

$\frac{(i)}{(ii)} \Rightarrow \frac{40}{20} = \log_{10}\left(\frac{I_1}{I_2}\right)$

$2 = \log_{10}\left(\frac{I_1}{I_2}\right)$ **or** $\frac{I_1}{I_2} = 10^2$

$\therefore \frac{r_2^2}{r_1^2} = 10^2$ (**since** $I\alpha\frac{1}{r^2}$)

$r_2^2 = 10^2 r_1^2$ **or** $r_2 = 10r_1 = 10 \times 1 = 10\ \mathrm{m}$

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